Introduction of Predicate calculus

The discussion of symbolic logic has been limited to the consideration of statements and statement formulae. It was not possible to express the fact that any two atomic statements have some features in common. In order to investigate questions of this nature, we introduce the concept of a predicate in an atomic statement. The logic based upon the analysis of predicates in any statement is called predicate logic.

Predicate calculus:

Let us consider two statements

- Ramu is clever
- Sita is clever

If we express these statements by symbols, we require two different symbols to denote them and it does not reveal the common features of these two statements i.e., both statements are about two different individuals who are clever.

We can introduce some symbol to denote " is clever" and a method to join it with symbols denoting the name of individuals. The part " is clever" is called a predicate.

We symbolize a predicate by a capital letter and the name of individuals or objects in general by small letters.

Let us denote:

Ramu by ‘r’

Sita by ‘s’

And a predicate “is clever” by a letter C

Now, the given statements can be written as $C(r)$ and $C(s)$.

Note: A statement which is expressed by using a predicate letter must have at least one name of object associated with the predicate.

Type of Predicates:

- One place predicate:

  This painting is red.
  Can be symbolized as R(p).

- Two place predicate:

  Ex: Mani is taller than Ravi.
  We can denote this as T(m,r).
  T is two place predicate as two subjects needed to complete the statement.

3-place predicate:

Ex: Mohan stands between Raj and Nitin.

B(m,r,n)

4-place predicate:

David & Tendulkar played cricket against Ganguly and Kambli.

P(d,t,g,k)

EXAMPLE 1

Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement "$x > 3$." Hence, $P(4)$, which is the statement "$4 > 3$," is true. However, $P(2)$, which is the statement "$2 > 3$," is false.

Statement function and variables

A simple statement function of one variable is defined to be an expression consisting of a predicate symbol and an individual variable.

Example: $M(x)$: x is a man.

We can also combine one or more statement functions using logical connectives which form compound statement function.

Example: $M(x)$: x is a man and $H(x)$: x is mortal, then $M(x) \wedge H(x)$, $M(x) \rightarrow H(x)$, $\sim M(x)$, etc

Same way we can define a statement function of two variables and more:

Ex: Let $G(x, y)$: x is taller than y.
If x is replaced by Pooja and y by Megha.

Then $G(x, y)$ becomes
Pooja is taller than Megha.

It is possible to form statement function of two variable from statement function of one variable:

Ex: M(x): x is a man.
H(y): y is mortal.

Then $M(x) \wedge H(y)$: x is a man and y is mortal.

Quantifiers

By which we can indicate the quantity by means of words like ‘all’, ‘some’, ‘none’, etc.

Example: ***Some*** man are mortal.

Types of quantifiers

1. Universal Quantifiers
2. Existential Quantifiers

Universal Quantifiers

The quantifiers 'all' is called the universal quantifier. It is denoted by the symbol $\forall \boldsymbol{x}$ or $(\boldsymbol{x})$ for the variable x.

Phrases for Universal Quantifiers:

1. For all x,
2. For every x,
3. For each x,
4. Everything x is such that
5. Each thing x is such that

Examples: "All men are mortal"

Let $M(x)$: x is a man

$H(x)$: x is Mortal.

The symbolic form of the given statement is $(\forall x)\big(M(x) \to H(x)\big)$.

Existential Quantifiers

The quantifiers 'some' is called the existential quantifier. It is denoted by the symbol $(\exists \boldsymbol{x})$ for some variable x.

Phrases for Existential Quantifiers:

1. For some x,
2. For some x such that,
3. There exist an x such that,
4. There is an x such that,
5. There is atleast one x such that.

Example 1: The proposition:

There is a dog without a tail can be written as

($\exists$ a dog) (the dog without tail)

Example 2: The proposition:

There is an integer between 2 and 8 inclusive may be written as

($\exists$ an integer) (the integer is between 2 and 8)

The propositions which include quantifiers may be negated as follows:

Example 3: Negate the proposition

All integers are greater than 8.

Solution: We can write the given proposition as

$(\forall \text{ integers } x)\ (x > 8)$

The negation is

$(\exists \text{ an integer } x)\ (x \leq 8)$

i.e., the negated proposition is: There is an integer less than or equal to 8.

In the negation a proposition ‘for all’ becomes ‘there is’ and ‘there is’ becomes ‘for all’ i.e., the symbol $\forall$ becomes $\exists$ and $\exists$ becomes $\forall$.

Example

Symbolize the following statements

a) Some men are honest.
b) Few cars have AC facility.
c) No cats has a tail.
d) All babies are innocent.

Universe of discourse

We can restrict our discussion to a particular set of objects or persons, (i.e.) the variables which are quantified stand for only those objects which are members of a particular set of class. Such a restricted class is called the universe of discourse or domain of discourse or universe.

Example: Let us take the statement

$$P(x):\ x \text{ is a cricket player}$$

then the domain of discourse for this statement can be taken as the set of all human beings

All cats are animals.

Which is true for any universe of discourse.
Let Universe of discourse is $E = \{cuddle, lilly, 0,1\}$,where cuddle and lilly are the names of the cat.

Then statement is true over E.

If we denote
C(x): x is a cat.
A(x): x is a animal.

Then the statement $(\forall x)\ (C(x) \rightarrow A(x))$ is true over E.

And the statement $(\forall x)$ (C$(x) \wedge A(x))$ is false.

i.e. in symbolic expressions of the type "All A are B" the correct connective that should be used is conditional.

Some cats are black.

Let Universe of discourse is $E = \{cuddle, lilly, kitty, 0, 1\}$,where cuddle and lilly are black cats.

If we denote
$C(x)$: x is a cat.
$B(x)$: x is a black.

Then the statement
$(\exists x)\ (C(x) \rightarrow B(x))$ is false over E.

And the statement $(\exists x)$ (C$(x) \wedge B(x)$) is true over E.

i.e. in symbolic expressions of the type "Some A are B" the correct connective that should be used as conjunction.

Free and Bound Variable

A part of the formula containing $(\forall x)P(x)$ or $(\exists x)\ P(x)$ is called x-bound part of the formula.

An occurrence of x in x bound part of a formula is called bound occurrence.

Any occurrence of x that is not bound occurrence is free occurrence.

Scope of the quantifier

The formula $P(x)$ in $(\forall x)P(x)$ or in $(\exists x)\ P(x)$ called scope of the quantifier i.e. scope of the quantifier is the formula immediately following the quantifier.

Example: $(\forall x)P(x,y)$
Here $P(x,y)$ is the scope of the quantifier.
Both occurrences of x is bound occurrences while occurrence of y is free occurrence.

Example: $(\forall x)(P(x) \rightarrow Q(x))$
Scope of the quantifier is $P(x) \rightarrow Q(x)$ and all occurrences of are bound occurrences.

Example 1. Find the truth value of $(\forall x)P(x)$, where $P(x){:}x^2 < 10$ and universe of discourse consist of a positive integer not exceeding 4.

Solution: The statement $(\forall x)P(x)$ for the given universe of discourse, is same as

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4) \Leftrightarrow T \wedge T \wedge T \wedge F$$
$$\Leftrightarrow \mathrm{F}$$

i.e. $(\forall x)P(x)$ is false.

- **Determine the truth value of following, consider set of real numbers as universe of discourse:**

$(\boldsymbol{a})\ \exists x,\ x^2 = x.$

Ans: true

$(b)\ \exists x,\ x + 2 = x.$

Ans: False

- **Determine the truth value if $A = \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \dots \dots, \mathbf{9}, \mathbf{10}\}$**

$(a)\ (x)(\exists y)(x + y < 14)$ Ans: True

(b) $(x)(y)(x + y < 14)$ Ans: False

Equivalence Formula of Predicate Calculus:

1. $(\exists x)\big(\mathrm{A(x)} \vee B(x)\big) \Leftrightarrow (\exists x)\mathrm{A(x)} \vee (\exists x)\mathrm{B(x)}$
2. $(\forall\ x)(A(x) \wedge B(x)) \Leftrightarrow (\forall\ x)A(x) \wedge (\forall\ x)B(x)$
3. $(\forall\ x)(A \vee B(x)) \Leftrightarrow A \vee (\forall\ x)B(x)$
4. $(\exists x)(A \wedge B(x)) \Leftrightarrow A \wedge (\exists x)B(x)$
5. $(\forall\ x)A(x) \to B \Leftrightarrow (\exists x)(A(x) \to B)$
6. $(\exists x)A(x) \to B \Leftrightarrow (\forall x)(A(x) \to B)$
7. $A \to (\forall\ x)B(x) \Leftrightarrow (\forall x)\big(A \to B(x)\big)$
8. $A \to (\exists x)B(x) \Leftrightarrow (\exists x)\big(A \to B(x)\big)$

Implications of Predicate Calculus

1. $(\forall x)\mathrm{A}(\mathrm{x}) \vee (\forall x)B(x) \Rightarrow (\forall x)\big(\mathrm{A}(\mathrm{x}) \vee \mathrm{B}(\mathrm{x})\big)$

2. $(\exists x)\big(\mathrm{A}(\mathrm{x}) \wedge B(x)\big) \Rightarrow (\exists x)\mathrm{A}(\mathrm{x}) \wedge (\exists x)\mathrm{B}(\mathrm{x})$

Demorgan's Laws for Predicate Calculus:

1. $\neg(\forall x)P(x) \Leftrightarrow (\exists x)\neg P(x)$
2. $\neg(\exists x)P(x) \Leftrightarrow (\forall x)\neg P(x)$

Proof : Let domain is the set S={$x_1, x_2, \ldots\ldots, x_n$}

Then $(\forall x)P(x) = P(x_1) \wedge P(x_2) \wedge \ldots\ldots \wedge P(x_n)$

And $(\exists x)P(x) = P(x_1) \vee P(x_2) \vee \ldots\ldots \vee P(x_n)$

Now, $\neg(\forall x)P(x) \Leftrightarrow \neg\,[P(x_1) \wedge P(x_2) \wedge \ldots\ldots \wedge P(x_n)]$

$\Leftrightarrow \neg P(x_1) \vee \neg P(x_2) \vee \ldots\ldots \vee \neg P(x_n)$

$\Leftrightarrow (\exists x)\neg P(x)$

i.e. $\neg(\forall x)P(x) \Leftrightarrow (\exists x)\neg P(x)$

Example : Symbolize the following statements and also negate them.

1. All cities in India are clean.
2. Some cities in India are polluted.
3. Some birds cannot fly.
4. No dog is Intelligent.
5. No cat has a tail.
6. All babies are innocent.
7. Few cars have AC facility.